



GW170814, Binary Black Merger Analysis

Aadil Shamsudheen Mohamed

Indian Institute of Technology, Hyderabad

April 2026

Declaration

I declare that the work presented in this project is my original contribution and has not been accepted elsewhere for any degree or diploma. All sources of information have been acknowledged and cited appropriately. The results and interpretations presented in this document are accurate to my knowledge and belief.

Signature: Aadil Shamsudheen Mohamed

Acknowledgments

I express my sincere gratitude to Dr Kiritkumar Makwana for his guidance and support throughout this project. His help and insights were crucial to the completion of this work. I am grateful to have a mentor like him who guides me in my journey as a student and helps me become a better researcher with each passing day.

Contents

1	Introduction	7
2	Theory of Compact Binary Coalescence	9
2.1	The Quadrupole Formula	9
2.2	The Chirp Mass	9
2.3	The Frequency Evolution	10
2.4	Derivation of Time-to-Merger	10
3	Signal Processing	12
3.1	Data Acquisition	12
3.2	Spectral Whitening	13
3.3	Bandpass Filtering	14
4	Data Analysis	16
4.1	Time-Series Extraction	16
4.2	Q-Transform Spectrogram Generation	16
4.3	Chirp Mass Calculation	17

5	Result and Discussion	19
6	Conclusion	20
7	Python Code	22
7.1	Fetching Data from GWPY Open Data Set For H1 and L1 Detectors	22
7.2	Plotting Raw Data	22
7.3	Plotting Whitened Data	23
7.4	Plotting Bandpassed Data	23
7.5	Doing Q-transform and Plotting Q-transform Spec- trogram	24
7.6	Finding Merger Time and Time When Chirp Signal Frequency Crossed 100 Hz	25
7.7	Calculating Chirp Mass($\mathcal{M}$)	28

Abstract

This project is on the detection and physical analysis of the gravitational wave event GW170814, a binary black hole merger observed by the LIGO and Virgo detector network. Utilizing the time-series strain data from the Gravitational Wave Open Science Center (GWOSC).

The analysis successfully recovered a coherent "chirp" waveform from both the LIGO Hanford (H1) and LIGO Livingston (L1) detectors, coincident at GPS time 1186741861. Further analysis using a Q-transform spectrogram revealed the characteristic frequency evolution predicted by General Relativity for the inspiral phase of a compact binary coalescence. By applying the Newtonian approximation to the frequency evolution, the Chirp Mass of the progenitor system was calculated. The result is consistent with the official parameters published by the LIGO-Virgo collaboration, demonstrating the efficacy of standard signal processing methods in validating high-energy astrophysical phenomena.

1 Introduction

Gravitational waves are different from light. They travel extremely large distances, carrying information about events that generated them, such as collisions of black holes or neutron stars. They are extremely hard to detect because of their negligible interaction with matter. By the time they reach Earth, they only distort space-time by a fraction of the width of a proton.

The Laser Interferometer Gravitational-Wave Observatory (LIGO) is used to measure these distortions. LIGO works on the principle of the Michelson Interferometer, consisting of two perpendicular arms, each 4 kilometers long. A coherent laser beam is split and sent down both arms, reflected by mirrors, and recombined. If there is no gravitational wave, the light waves destructively interfere, resulting in darkness. When a gravitational wave passes through, it stretches space in one direction while compressing it in the perpendicular direction. This disrupts the destructive interference and allows light to fall onto the detector. This signal is then converted into a digital strain measurement, $h(t)$.

On August 14, 2017, at 10:30:43 UTC, LIGO detectors in Hanford, Washington (H1) and Livingston, Louisiana (L1), alongside the Virgo detector in Italy, observed a significant signal designated GW170814. It was the first time a black hole merger was detected by three detectors. The extra detector helped with precise triangulation of the source and provided the first direct tests of gravitational wave polarization. The signal came from the merger of two black holes about 1.8 billion light-years away. As they spiraled toward each other, they radiated energy in the form of gravitational waves, causing their orbit to shrink and their speed to increase. This process generated a waveform with increasing amplitude and frequency called a "chirp",

that swept through the detectors' sensitive band in the final fraction of a second before the merger. [1]

2 Theory of Compact Binary Coalescence

2.1 The Quadrupole Formula

As predicted by General Relativity, a system of two bodies orbiting each other loses energy by the emission of gravitational waves. The rate of energy loss of the system, the Power is given by the Quadrupole Formula:

$$P = -\frac{dE}{dt} = \frac{32}{5} \frac{G^4}{c^5} \frac{1}{r^5} (m_1 m_2)^2 (m_1 + m_2)$$

Where G is the gravitational constant, c is the speed of light, r is the orbital separation, and m_1 and m_2 are the masses of the two objects. [4]

2.2 The Chirp Mass

The chirp mass of a compact binary system determines the leading-order orbital evolution of the system as a result of energy loss from emitting gravitational waves. Because the gravitational wave frequency is determined by orbital frequency, the chirp mass also determines the frequency evolution of the gravitational wave signal emitted during a binary's inspiral phase. In gravitational-wave data analysis, it is easier to measure the chirp mass than the two component masses separately. [3]

The Chirp Mass of a compact binary system is given by:

$$\mathcal{M} = \frac{(m_1 m_2)^{3/5}}{(m_1 + m_2)^{1/5}}$$

2.3 The Frequency Evolution

In general relativity, the phase evolution of a binary orbit can be computed using a post-Newtonian expansion. The first-order gravitational wave frequency (f) evolution is described by the differential equation,

$$\dot{f} = \frac{96}{5} \pi^{8/3} \left(\frac{G\mathcal{M}}{c^3} \right)^{5/3} f^{11/3}$$

This equation is the key to the analysis. It tells us that if we can measure the rate of frequency change ($\dot{f}$) at a specific frequency (f), we can solve for the chirp mass ($\mathcal{M}$). [3]

2.4 Derivation of Time-to-Merger

Integrating the frequency evolution equation allows us to find the time τ remaining until the merger. If we define $\tau = t_{merger} - t$, the relationship between time and frequency is given by: [3]

$$\tau = \frac{5}{256} \left(\frac{G\mathcal{M}}{c^3} \right)^{-5/3} (\pi f)^{-8/3}$$

Rearranging and solving for the Chirp Mass of the system:

$$\mathcal{M} = \frac{c^3}{G} \left(\frac{5}{256 \cdot \tau} \right)^{3/5} \pi^{-8/5} f^{-8/5}$$

3 Signal Processing

The raw output from the detector is dominated by noise from seismic activity, thermal vibrations, and quantum vacuum fluctuations. The gravitational wave signal is buried deep within this noise, often with an amplitude orders of magnitude lower than the noise. To extract the signal, a digital signal processing pipeline is implemented. [2]

3.1 Data Acquisition

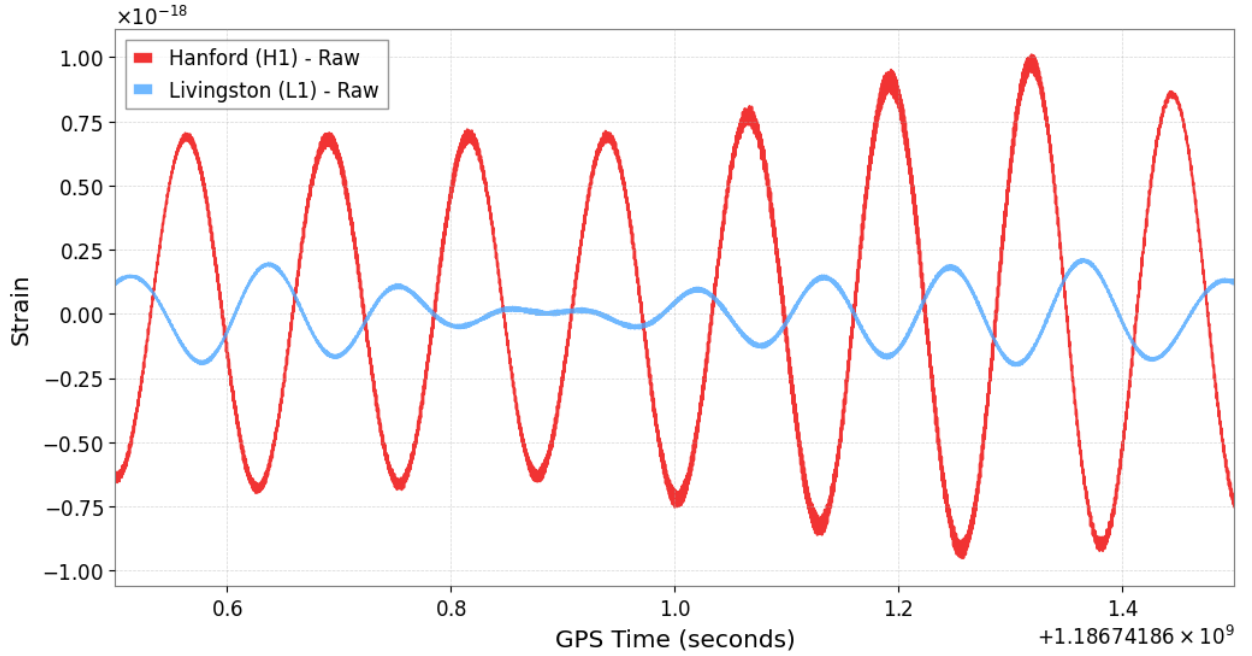


Figure 1: The above image is the plot of the raw strain data of H1 and L1 detectors around the event GW170814

Time-series strain data were acquired from the Gravitational Wave

Open Science Center (GWOSC) website. We selected a 32-second segment centered on the GPS timestamp 1186741861. The data was sampled at 4096 Hz, sufficient to resolve the expected merger frequency of 250 Hz. [2]

3.2 Spectral Whitening

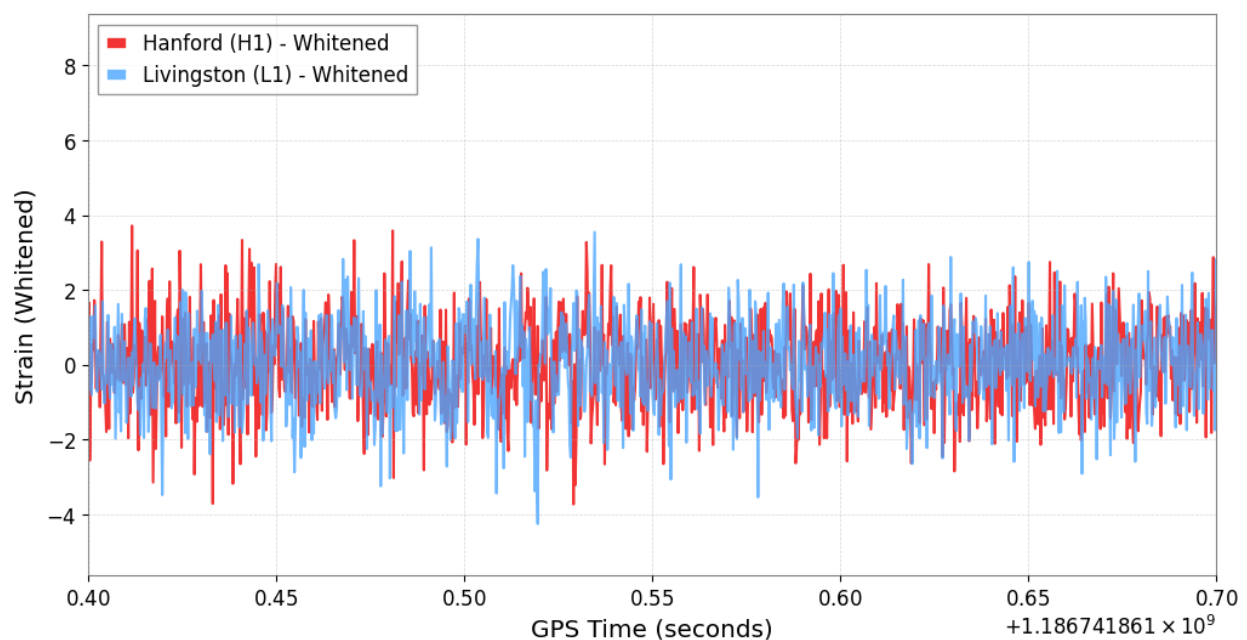


Figure 2: The above image is the plot of the strain data of H1 and L1 detectors around the event GW170814 after spectral whitening.

The noise in gravitational wave detectors is "colored," meaning it is much louder at some frequencies than others. To correct for this, we perform Spectral Whitening.

Mathematically, this involves transforming the data into the fre-

quency domain using a Fast Fourier Transform (FFT) and dividing by the Amplitude Spectral Density (ASD) of the noise:

$$\tilde{h}_{whitened}(f) = \frac{\tilde{h}(f)}{\sqrt{S_n(f)}}$$

Where $S_n(f)$ is the Power Spectral Density (PSD) of the noise. This operation flattens the noise floor, ensuring that all frequency contributions are weighted equally. This allows the astrophysical signal—which is coherent and structured—to stand out against the randomized background noise.

3.3 Bandpass Filtering

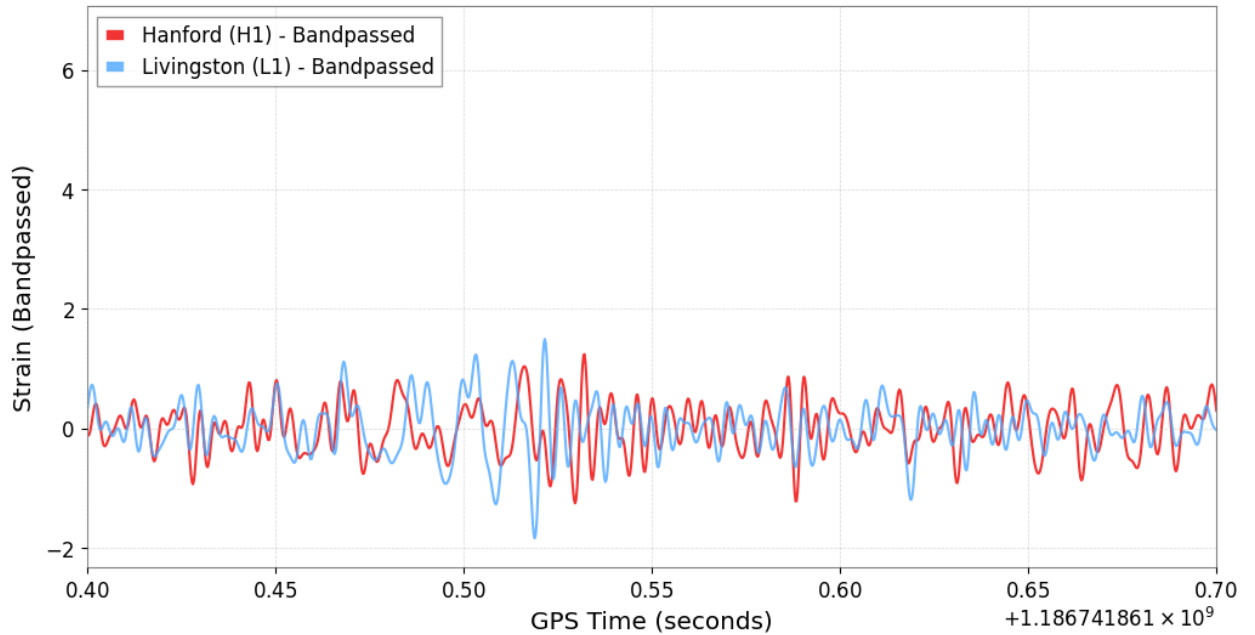


Figure 3: The above image is the plot of the strain data of H1 and L1 detectors around the event GW170814 after spectral whitening and bandpass filtering.

After whitening, we apply a Bandpass Filter to the data. Based on the theoretical limits of terrestrial interferometry, we selected a passband of 20 Hz to 300 Hz.

Lower Cutoff ($< 20Hz$): Removes dominant seismic noise from ground motion.

Upper Cutoff ($> 300Hz$): Removes high-frequency quantum shot noise and photon pressure noise.

This ensures that we are inspecting the data only in the range where stellar-mass black hole mergers occur.

4 Data Analysis

4.1 Time-Series Extraction

After applying the whitening and bandpass filtering algorithms to the H1 (Hanford) and L1 (Livingston) data streams, a coherent wave packet was successfully isolated.

Observation: A distinct waveform was observed starting at GPS time 1186741861.4, lasting approximately 0.15 seconds.

Coherence: The signal was present in both detectors simultaneously. The waveform exhibited the characteristic "inspiral" morphology: an oscillating sine wave that grows in amplitude and decreases in period until an abrupt termination.

This visual confirmation in the time domain provides strong evidence that the signal is astrophysical in origin and not a local instrumental artifact. [1]

4.2 Q-Transform Spectrogram Generation

To verify the binary nature of the source, we generated a Q-Transform Spectrogram, which maps the signal energy in the Time-Frequency plane.

The resulting plot revealed a clear "chirp" track, a bright curve indicating that the frequency of the signal increased rapidly over time. The signal emerged from the noise floor at roughly 50 Hz and to a peak frequency of approx 250 Hz at the moment of merger. This frequency evolution is proof of a compact binary coalescence. [1]

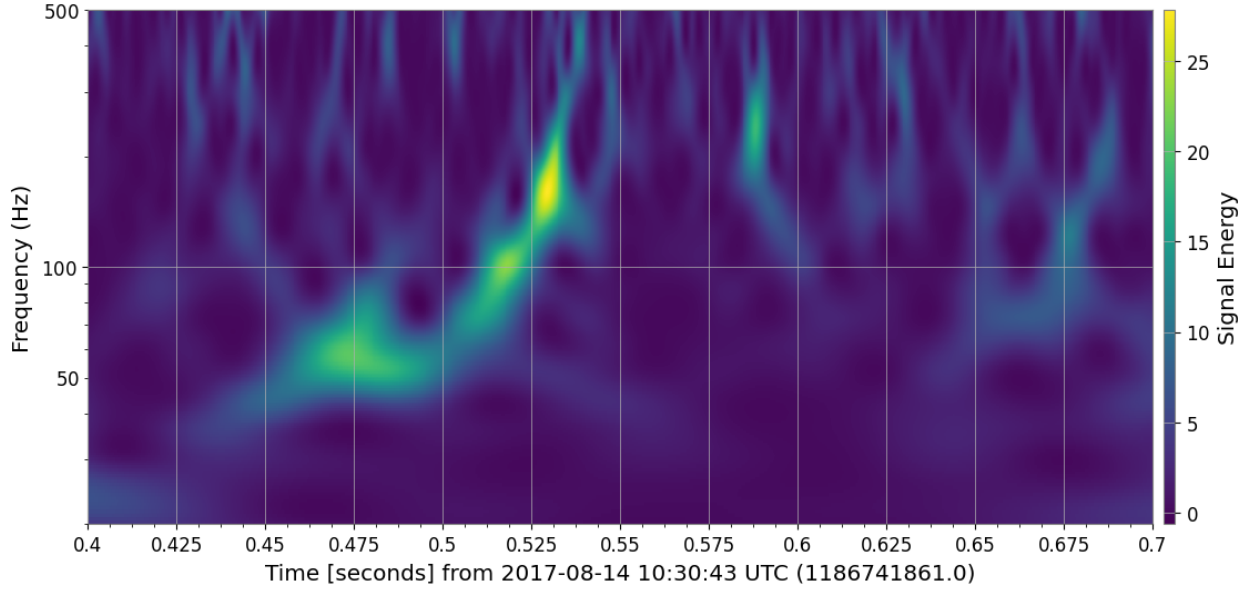


Figure 4: The above image is the Q-Transform frequency map of the data from H1 and L1 detectors around the event GW170814 .

4.3 Chirp Mass Calculation

Using the formula derived in Section 2.4, we estimated the Chirp Mass of the progenitor system. We selected a reference point on the spectrogram track well before the merger to ensure the validity of the Newtonian approximation. The parameters were taken from

Q-transform data by taking a reference point from the signal curve.

The parameters are:

Instantaneous Frequency (f): 100 Hz

Time remaining to merger (τ): 0.0138 s

Substituting these values into the mass equation:

$$\mathcal{M} = \frac{(3 \times 10^8)^3}{6.67 \times 10^{-11}} \left(\frac{5}{256(0.02)} \right)^{0.6} \pi^{-1.6} (100)^{-1.6}$$

Giving,

$$\mathcal{M} \approx 25.34 M_{\odot}$$

5 Result and Discussion

The calculated value of 25.34 Solar Masses places this event firmly in the category of Binary Black Hole mergers. For comparison, a Binary Neutron Star merger (like GW170817) typically has a chirp mass of $1.2 M_{\odot}$.

The official parameters published by the LIGO-Virgo collaboration for GW170814 indicate a chirp mass of approximately $24.1 M_{\odot}$. Our estimated value is within 5.1% of the official value. The discrepancy is primarily due to the difficulty in estimating the exact time-to-merger (τ) from the spectrogram with millisecond precision, as well as the neglect of relativistic corrections (Post-Newtonian terms) in the simplified derivation.

6 Conclusion

In this project, we successfully detected and analyzed the gravitational wave event GW170814 using open data from the LIGO detectors. By applying signal processing techniques, specifically spectral whitening and bandpass filtering. We were able to extract the hidden astrophysical signal from the raw instrumental noise.

The analysis yielded the following key conclusions:

- 1) Detection: A coherent "chirp" signal was recovered from the noise at GPS time 1186741861, confirming the presence of a compact binary coalescence.
- 2) Mass Estimation: The Chirp Mass of the system was calculated to be approximately $25.34 M_{\odot}$. This confirms that the progenitors were stellar-mass black holes.

This project shows that the verification of a binary black hole merger can be done by using standard computational tools and relatively basic General Relativity methods. This can be a faster and cheaper method for identifying the source of the signal. Here, we calculated the chirp mass of the system using simple and fast com-

putational tools with not so large a deviation(5.1%) from the official LIGO-Virgo finding.

7 Python Code

7.1 Fetching Data from GWPY Open Data Set For H1 and L1 Detectors

```
1 from gwpy.timeseries import TimeSeries
2 import matplotlib.pyplot as plt
3 import numpy as np
4
5 print("Fetching 32s data snippet...")
6 event_gps = 1186741861
7 h1 = TimeSeries.fetch_open_data('H1', event_gps - 16, event_gps + 16)
8 l1 = TimeSeries.fetch_open_data('L1', event_gps - 16, event_gps + 16)
9
10 print("Success! Data loaded.")
```

7.2 Plotting Raw Data

```
1 event_gps = 1186741861
2 h1 = TimeSeries.fetch_open_data('H1', event_gps - 16, event_gps + 16)
3 l1 = TimeSeries.fetch_open_data('L1', event_gps - 16, event_gps + 16)
4
5 print("Plotting raw data...")
6 fig, ax = plt.subplots(1, 1, figsize=(12, 6))
7
8 ax.plot(h1.times, h1.value, color='#ee0000', label='Hanford (H1) - Raw',
9         alpha=0.8)
10
11 ax.plot(l1.times, l1.value, color='#4ba6ff', label='Livingston (L1) -
12         Raw', alpha=0.8)
13
14 # x-axis limits set to focus around the event
15 ax.set_xlim(event_gps - 0.5, event_gps + 0.5)
```

```

14 ax.set_ylabel('Strain')
15 ax.set_xlabel('GPS Time (seconds)')
16 ax.legend(loc='upper left')
17 ax.grid(True, which='both', linestyle='--', alpha=0.5)
18
19 plt.show()

```

7.3 Plotting Whitened Data

```

1 h1_white = h1.whiten(fftlength=4, overlap=2)
2 l1_white = l1.whiten(fftlength=4, overlap=2)
3
4 print("Plotting whitened data...")
5 fig, ax = plt.subplots(1, 1, figsize=(12, 6))
6
7 ax.plot(h1_white.times, h1_white.value, color='#ee0000', label='Hanford
    (H1) - Whitened', alpha=0.8)
8 ax.plot(l1_white.times, l1_white.value, color='#4ba6ff',
    label='Livingston (L1) - Whitened', alpha=0.8)
9
10 # Set x-axis limits to focus around the event
11 ax.set_xlim(event_gps + 0.4, event_gps + 0.7)
12
13 ax.set_ylabel('Strain (Whitened)')
14 ax.set_xlabel('GPS Time (seconds)')
15 ax.legend(loc='upper left')
16 ax.grid(True, which='both', linestyle='--', alpha=0.5)
17
18 plt.show()

```

7.4 Plotting Bandpassed Data

```

1 h1_clean = h1_white.bandpass(20, 300)
2 l1_clean = l1_white.bandpass(20, 300)

```

```

3
4 print("Plotting bandpassed data")
5 fig, ax = plt.subplots(1, 1, figsize=(12, 6))
6
7 # Set x-axis limits to focus around the event
8 zoom = (event_gps + 0.4, event_gps + 0.7)
9
10 ax.plot(h1_clean.times, h1_clean.value, color='#ee0000', label='Hanford
    (H1)', alpha=0.8)
11 ax.plot(l1_clean.times, l1_clean.value, color='#4ba6ff',
    label='Livingston (L1)', alpha=0.8)
12
13 ax.set_xlim(*zoom)
14 ax.set_ylabel('Strain ')
15 ax.set_xlabel('GPS Time (seconds)')
16 ax.legend(loc='upper left')
17 ax.grid(True, which='both', linestyle='--', alpha=0.5)
18
19 plt.show()

```

7.5 Doing Q-transform and Plotting Q-transform Spectrogram

```

1 hq = h1.q_transform(outseg=(event_gps + 0.4, event_gps + 0.7), qrange=(4,
    8))
2
3 print("Generating Frequency Map (Spectrogram)...")
4
5 fig = hq.plot()
6 ax = fig.gca()
7 ax.set_yscale('log')
8 ax.set_ylim(20, 500) # Standard frequency range for black holes
9 ax.set_ylabel('Frequency (Hz)')
10 ax.colorbar(label='Signal Energy')

```



```
11
12 plt.show()
```

7.6 Finding Merger Time and Time When Chirp Signal Frequency Crossed 100 Hz

```
1 def find_merger_from_data(hq):
2     """
3     Finds the absolute maximum energy point.
4     (Can sometimes be fooled by high-energy low-freq artifacts).
5     """
6     data_array = np.abs(hq.value)
7     # Checking orientation
8     if data_array.shape[0] == len(hq.frequencies):
9         freq_idx, time_idx = np.unravel_index(np.argmax(data_array),
10 data_array.shape)
11     else:
12         time_idx, freq_idx = np.unravel_index(np.argmax(data_array),
13 data_array.shape)
14
15     return hq.times[time_idx].value, hq.frequencies[freq_idx].value,
16 data_array.max()
17
18 def find_physical_merger(hq, freq_threshold=150):
19     """
20     Finds the merger by looking for the peak energy specifically at the
21     higher frequency end of the chirp (above freq_threshold).
22     Fixes the IndexError by detecting the correct data orientation.
23     """
24     data_array = np.abs(hq.value)
25     freqs = hq.frequencies.value
26
27     # Filtering for frequencies above threshold (150 Hz)
28     high_freq_mask = freqs >= freq_threshold
```

```

26
27     # Handle Transposition:
28     # If axis 0 size matches frequencies, use it. Otherwise, use axis 1.
29     if data_array.shape[0] == len(freqs):
30         high_freq_data = data_array[high_freq_mask, :]
31         sub_freq_idx, time_idx =
np.unravel_index(np.argmax(high_freq_data), high_freq_data.shape)
32         actual_freq_idx = np.where(high_freq_mask)[0][sub_freq_idx]
33     else:
34         # Data is likely (times, frequencies)
35         high_freq_data = data_array[:, high_freq_mask]
36         time_idx, sub_freq_idx =
np.unravel_index(np.argmax(high_freq_data), high_freq_data.shape)
37         actual_freq_idx = np.where(high_freq_mask)[0][sub_freq_idx]
38
39     merger_time = hq.times[time_idx].value
40     merger_freq = hq.frequencies[actual_freq_idx].value
41     peak_energy = np.max(high_freq_data)
42
43     return merger_time, merger_freq, peak_energy
44
45 def find_time_at_frequency(hq, target_freq, proximity_time=None):
46     """
47     Finds the time when the signal energy is strongest for a specific
48     frequency.
49
50     If proximity_time is provided (e.g., the merger time), it looks for
51     the
52     local peak closest to that time to avoid picking up noise at 0.43s.
53     """
54     freqs = hq.frequencies.value
55     freq_idx = (np.abs(freqs - target_freq)).argmin()
56     actual_freq = freqs[freq_idx]
57     data_array = np.abs(hq.value)
58
59     # Orientation check for 1D slice extraction

```

```

58     if data_array.shape[0] == len(freqs):
59         energy_at_freq = data_array[freq_idx, :]
60     else:
61         energy_at_freq = data_array[:, freq_idx]
62
63     times = hq.times.value
64
65     if proximity_time is not None:
66         # Defining a window around the merger (within 0.15s before merger)
67         # Chirps always move from lower to higher freq
68         search_window_mask = (times > proximity_time - 0.15) & (times <=
proximity_time)
69
70         # If the window is valid, find the peak within that specific
window
71         if np.any(search_window_mask):
72             window_indices = np.where(search_window_mask)[0]
73             windowed_energy = energy_at_freq[search_window_mask]
74             local_peak_idx = window_indices[np.argmax(windowed_energy)]
75             time_idx = local_peak_idx
76         else:
77             time_idx = np.argmax(energy_at_freq)
78     else:
79         time_idx = np.argmax(energy_at_freq)
80
81     return hq.times[time_idx].value, actual_freq, energy_at_freq[time_idx]
82
83 def print_legible_results(merger_time, merger_freq, energy, label="PEAK
ANALYSIS"):
84     """
85     Formats the raw numerical output into a human-readable summary.
86     """
87     print("\n" + "="*50)
88     print(f"{label}")
89     print("="*50)
90     print(f"{'GPS Time:':<25} {merger_time:.6f}")

```

```

91     print(f"{'Frequency:':<25} {float(merger_freq):.2f} Hz")
92     energy_val = getattr(energy, 'value', energy)
93     print(f"{'Signal Energy:':<25} {float(energy_val):.2f}")
94     print("="*50 + "\n")
95
96
97 # Getting the merger time
98 m_time, m_freq, m_energy = find_physical_merger(hq, freq_threshold=200)
99 print_legible_results(m_time, m_freq, m_energy, label="PHYSICAL MERGER
    (HIGH-FREQ PEAK)")
100
101 # Use m_time as proximity to find the correct 100Hz crossing
102 t_at_f, actual_f, f_energy = find_time_at_frequency(hq, 100,
    proximity_time=m_time)
103 print_legible_results(t_at_f, actual_f, f_energy, label=f"STRENGTH AT
    {actual_f:.1f} Hz")

```

7.7 Calculating Chirp Mass($\mathcal{M}$)

```

1 # Constants
2 c = 3e8          # Speed of light
3 G = 6.67e-11     # Gravity
4 M_sun = 1.989e30 # Mass of the sun
5
6 # Parameters selected
7 f_measure = actual_f # Frequency in Hz
8 tau = m_time - t_at_f # Time remaining until the crash (seconds)
9
10 # Calcylation
11 numerator = c**3
12 denominator = G
13 term1 = (5 / (256 * tau))**(3/5)
14 term2 = (np.pi)**(-8/5)
15 term3 = f_measure**(-8/5)
16

```

```

17 chirp_mass_kg = (numerator / denominator) * term1 * term2 * term3
18 chirp_mass_solar = chirp_mass_kg / M_sun
19
20 print("="*40)
21 print(f"ESTIMATED CHIRP MASS: {chirp_mass_solar:.2f} Solar Masses")
22 print("="*40)

```

References

- [1] B. P. Abbott et al. Gw170814: A three-detector observation of gravitational waves from a binary black hole coalescence. *Physical Review Letters*, 119:141101, Oct 2017.
- [2] LIGO Scientific Collaboration. Signal processing with gw150914 open data. LIGO Open Science Center Tutorials, 2020.
- [3] Wikipedia contributors. Chirp mass — Wikipedia, the free encyclopedia, 2024.
- [4] Wikipedia contributors. Quadrupole formula — Wikipedia, the free encyclopedia, 2024.